



DEMOCRATIC AND POPULAR REPUBLIC OF ALGERIA

UNIVERSITY OF SAAD DAHLEB BLIDA 1

Faculty of Science Department of Computer
Science



Presented for obtaining the Master's degree In computer science

Sentiment analysis in Arabic Dialects

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1. Problematique & Background

Imagine a company launching a new product and wanting to know how customers feel about it on social media.

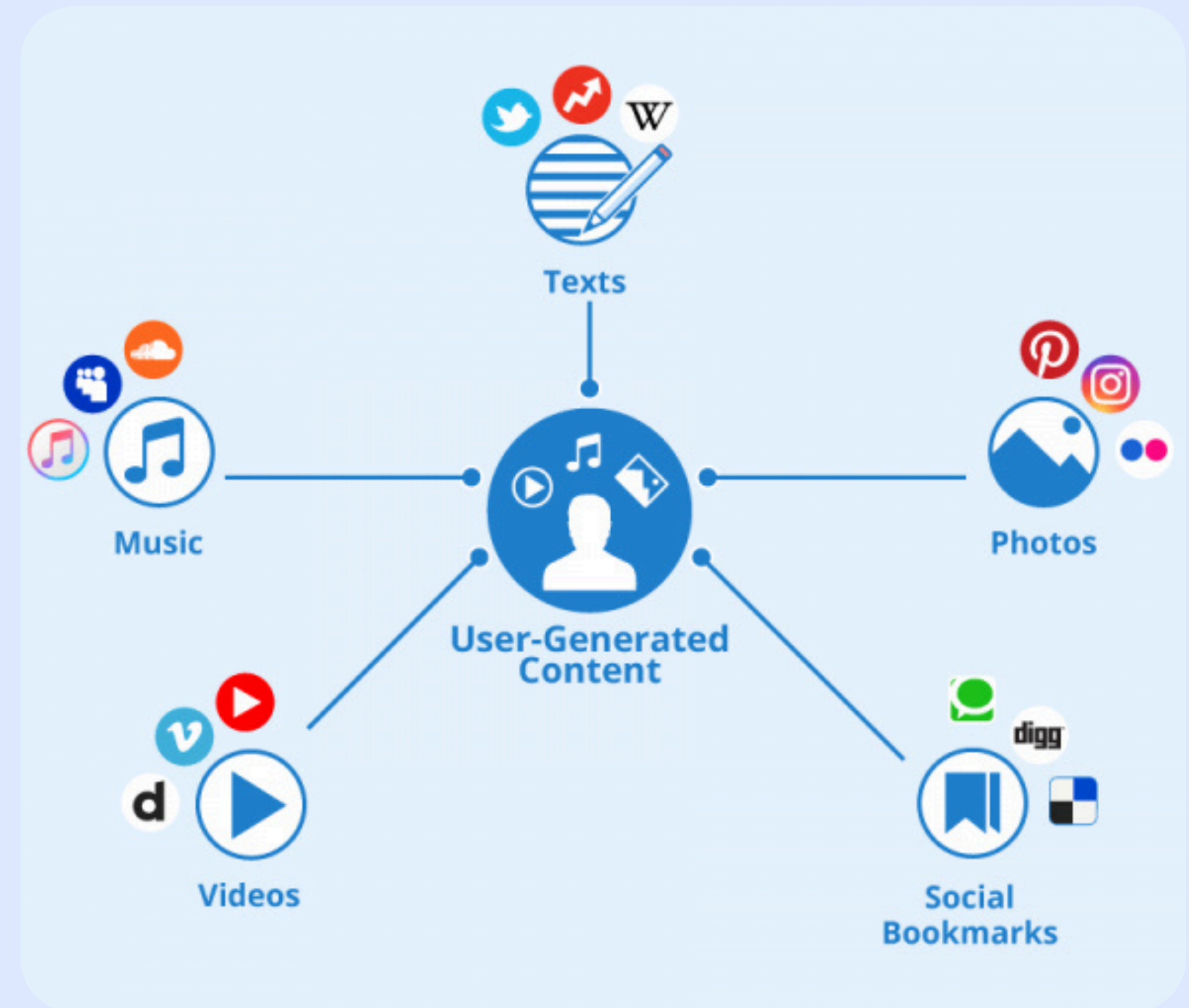


10.000 comment

But how can a company quickly understand this huge volume of feedback and make better decisions?

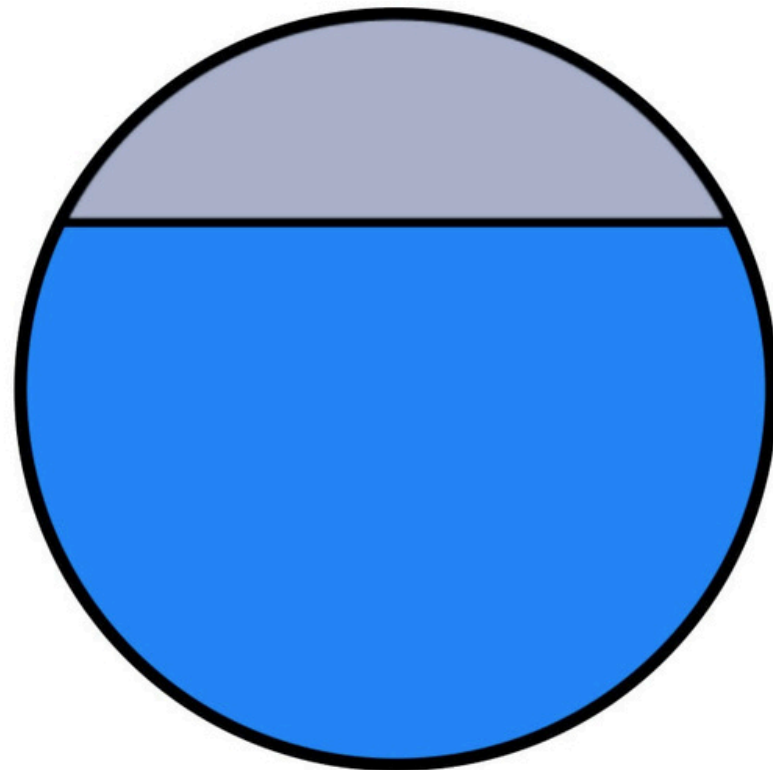
What is User Generated Content (UGC) ?

All of this feedback—tweets, comments, reviews, forum posts, and even YouTube videos—is part of something called user-generated content.



What is User Generated Content (UGC) ?

In 2023, 75% of all content online is UGC



■ UGC
■ NON-UGC

2023 report by Statista

YouTube Statistics

Over 500 hours of content are uploaded to YouTube every minute.

500 Hours Uploaded Every Minute

analysify.com

UGC statistics

Benefits and impact on business success.



92% of consumers **trust UGC more** than traditional advertising.



79% of consumers say **UGC highly impacts** their purchasing decisions

embedsocial.com

Sentiment analysis

Sentiment analysis is a powerful tool that automatically detects whether a piece of text expresses a positive, negative, or neutral opinion.



My Experience
so far has been
fantastic!

Positive



The product
is ok I guess

Neutral



Your support
team is useless

Negative

Use case of Sentiment analysis

How would
sentiment analysis
be applied?



Brand monitoring

Track sentiment to manage brand reputation and influence public perception



Customer service

Prioritize negative sentiment to enhance customer satisfaction and loyalty

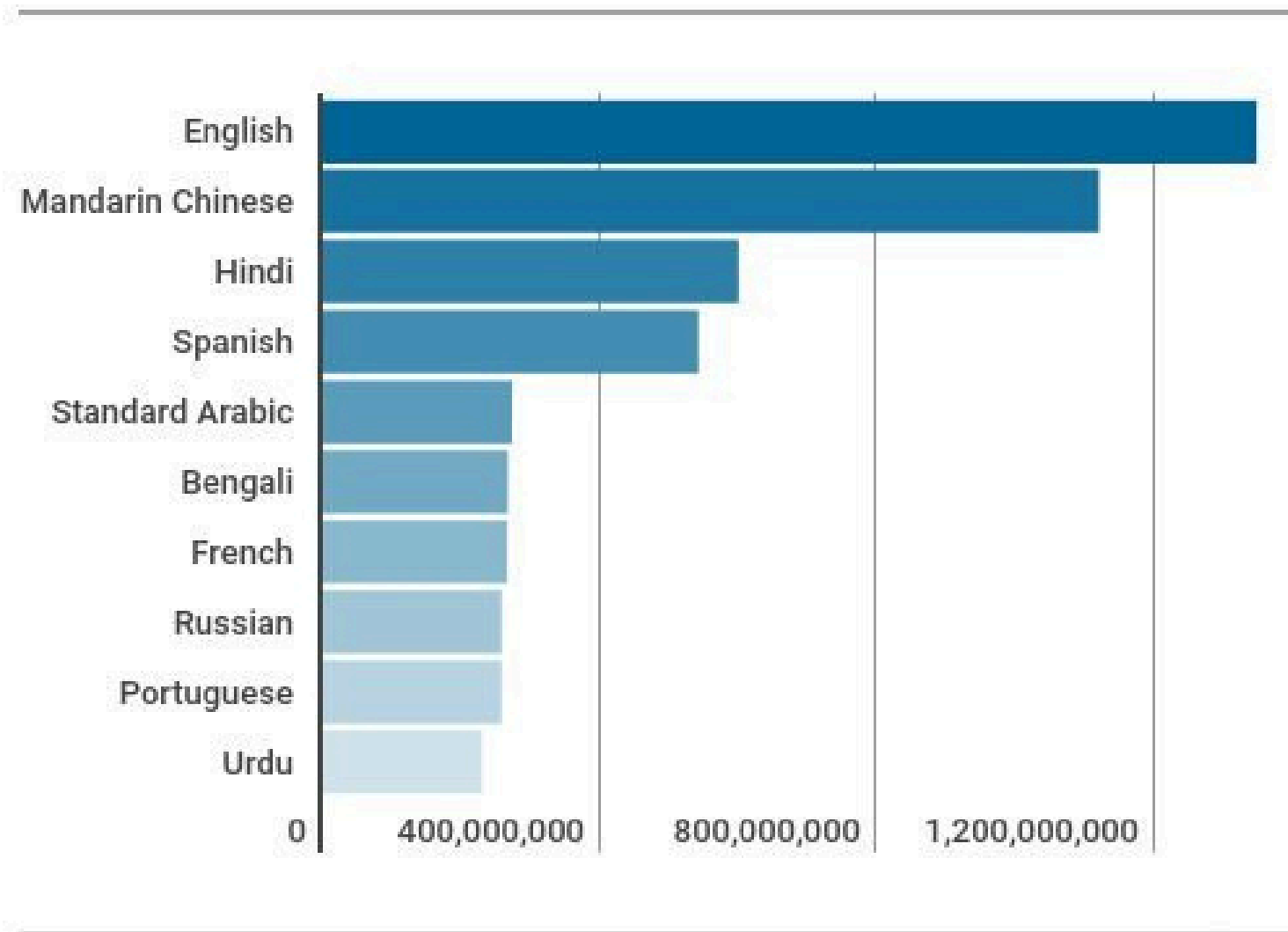


Market Research

Analyze sentiment towards competitors to identify market opportunities

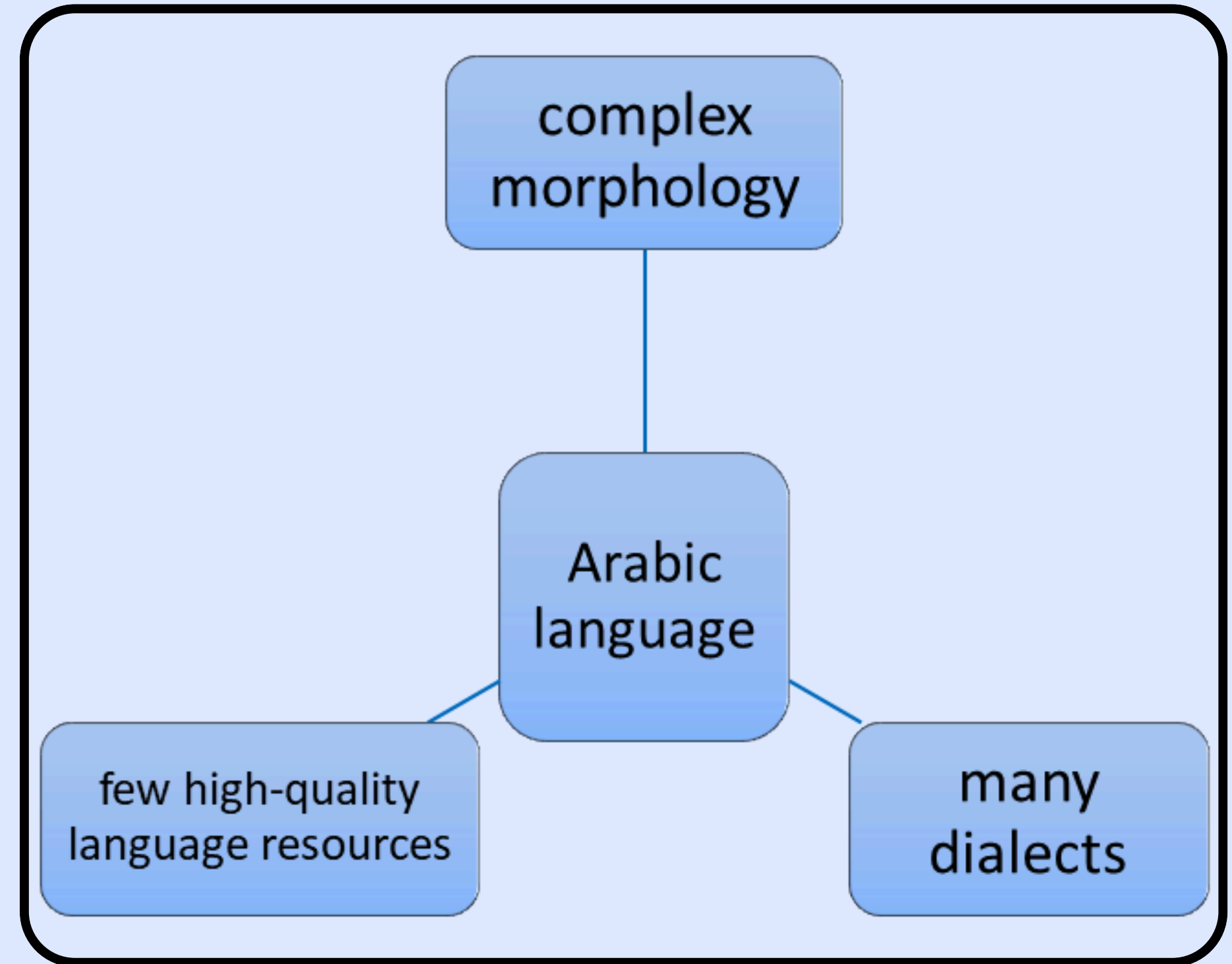
Challenges of Arabic Language

Top 10 most spoken languages, 2021



Ethnologue

ethnologue.com 2021

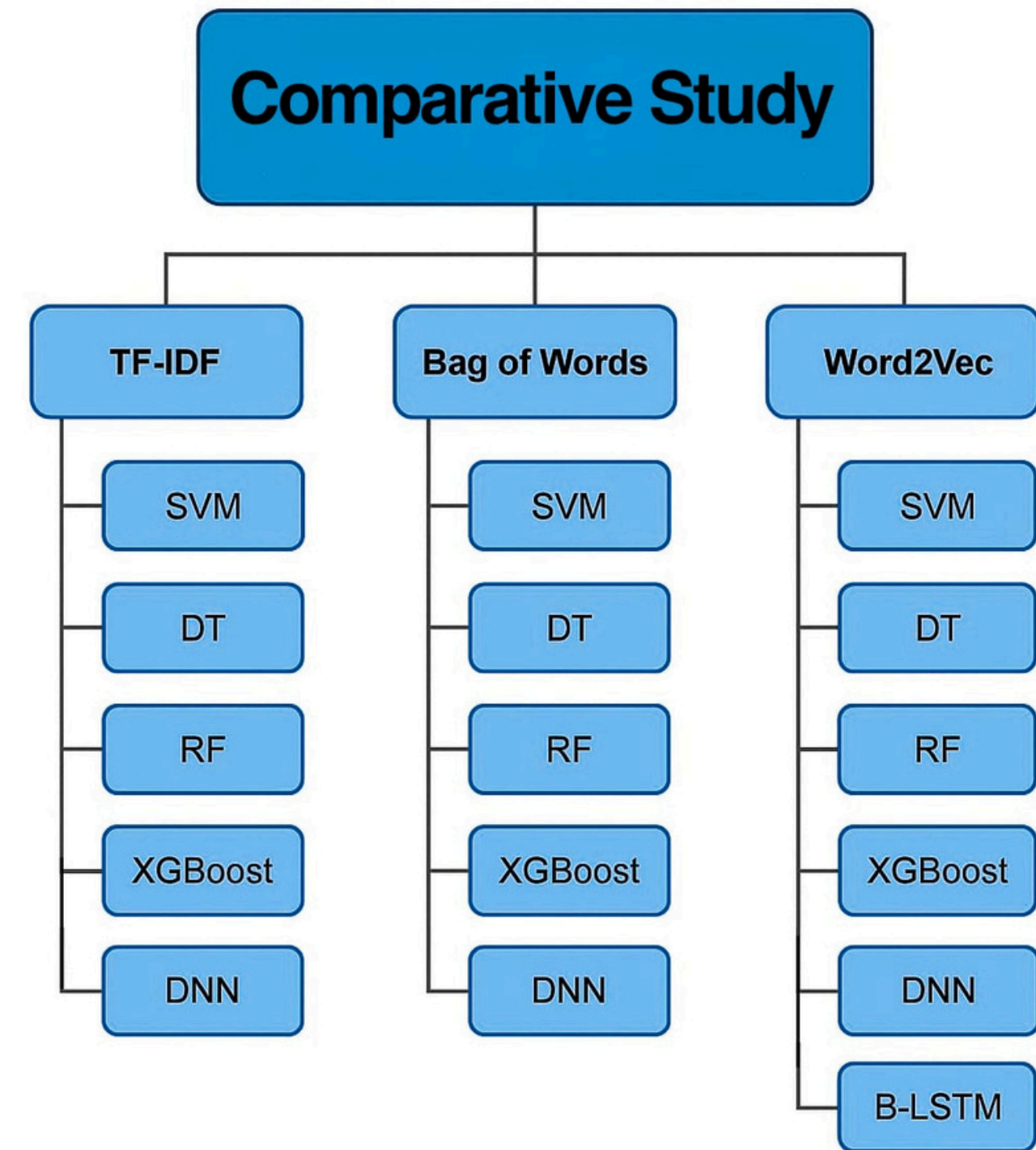


challenges for arabic sentiment analysis

2-Research Objectives

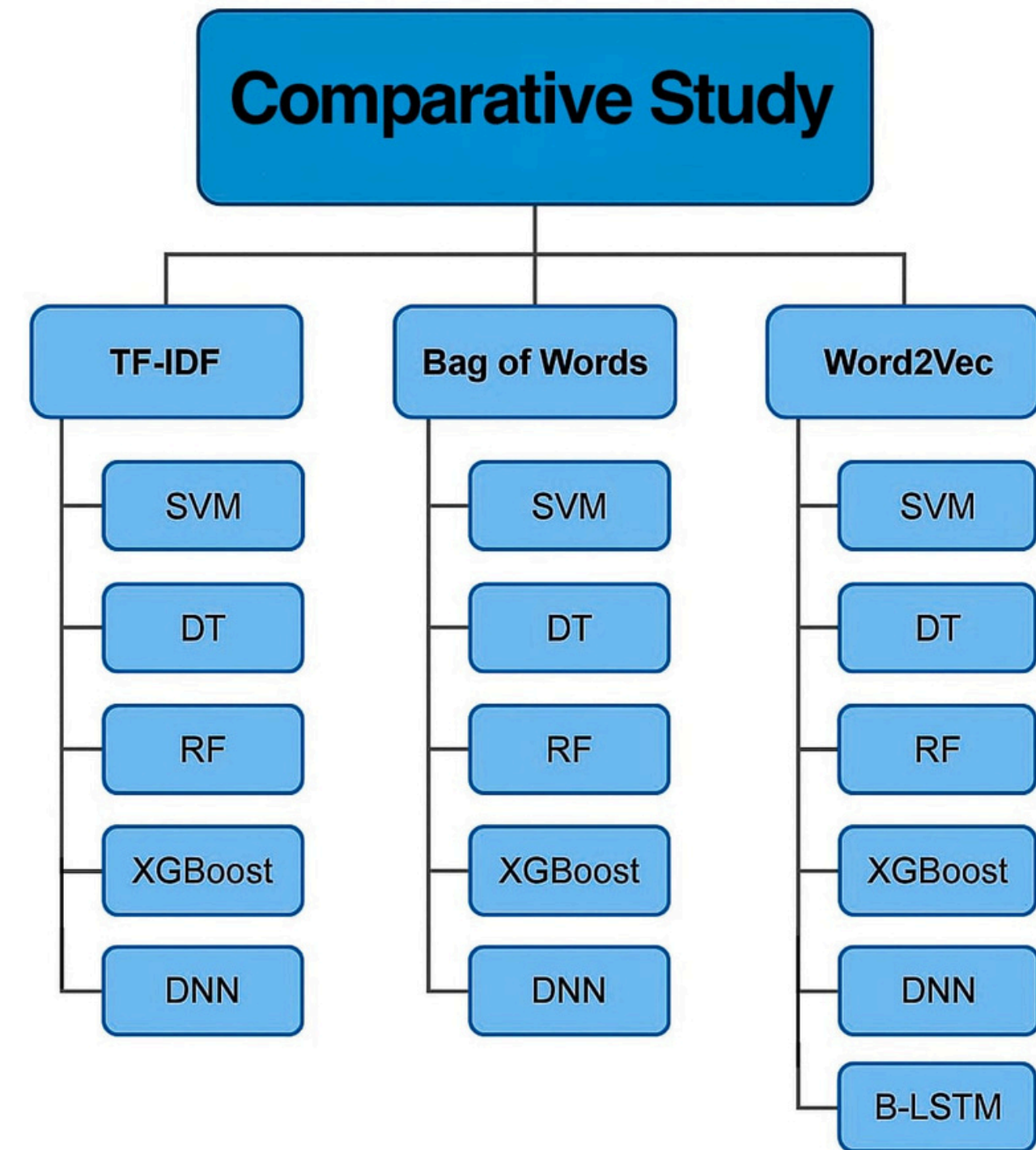
The main goal of our work is to conduct a comprehensive comparative study of sentiment analysis techniques for Arabic text.

Specifically, we aim to evaluate and compare different text representation methods and a variety of machine learning and deep learning models.

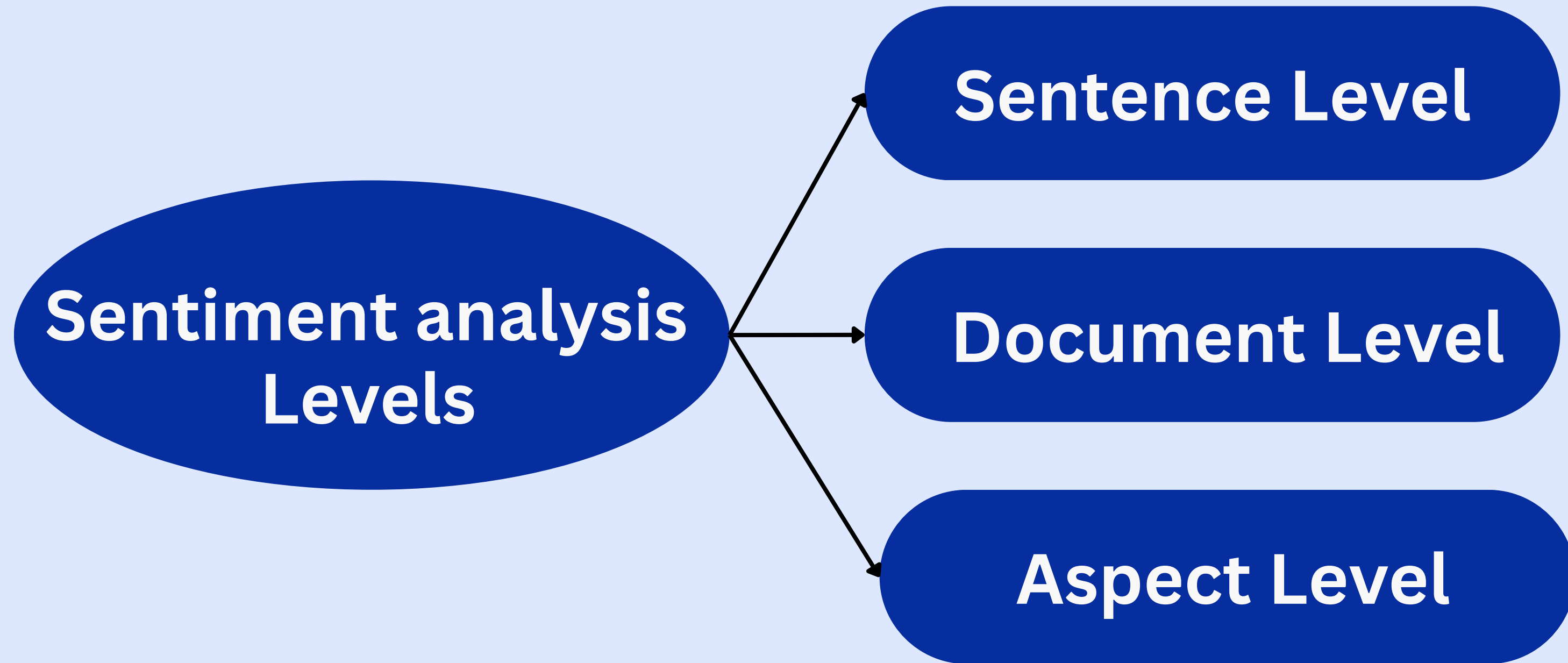


2-Research Objectives

By systematically comparing these approaches, our objective is to provide clear, evidence-based recommendations for building effective sentiment analysis systems designed to the unique challenges of the Arabic language.



3-Related Works



3-Related Works

Paper	Encoding	Models / Techniques	Results
Titya Eng et al. (2016)	TF-IDF	Lexicon + ML	Outperforms baseline ML
Rukhsar et al. (2023)	TF-IDF for SVM,Word Embeddings for LSTM	LSTM, SVM	LSTM: 87%, SVM: 86% (Binary: both 90%)
Behdenna et al. (2018)	BoW, TF-IDF	SVM, NB, Lexicon, CNN, LSTM	Hybrid (NB + GA) gave the best accuracy
Maryem Rhanoui et al. (2019)	Doc2Vec	CNN, LSTM, CNN-BiLSTM	CNN-BiLSTM + Doc2Vec → 90.66% accuracy 12

3-Related Works

Paper	Embedding	Models / Techniques	Results
Bhoite & Londhe (2021)	BoW	Naïve Bayes	Precision, recall, F-score used; accuracy not specified
Hai Ha Do et al. (2019)	Word embeddings (Word2Vec, etc.)	DL: LSTM, Attention, GRU, etc.	Comparative review; shows DL > ML for aspect analysis

4. Dataset & Preprocessing

1

Dataset
description

2

Data Cleaning

3

Tokenization

4

Label
Encoding

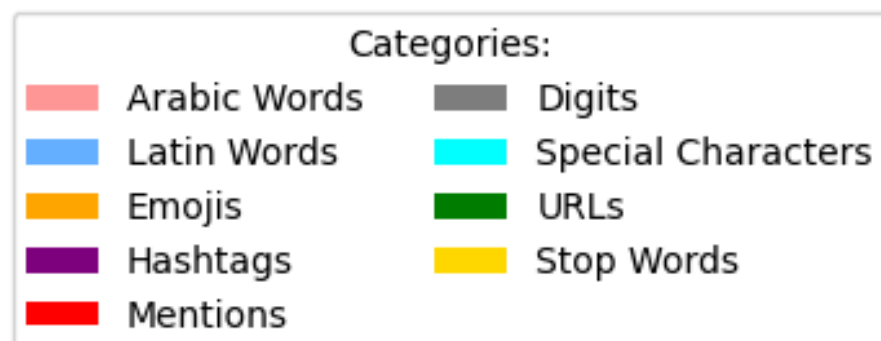
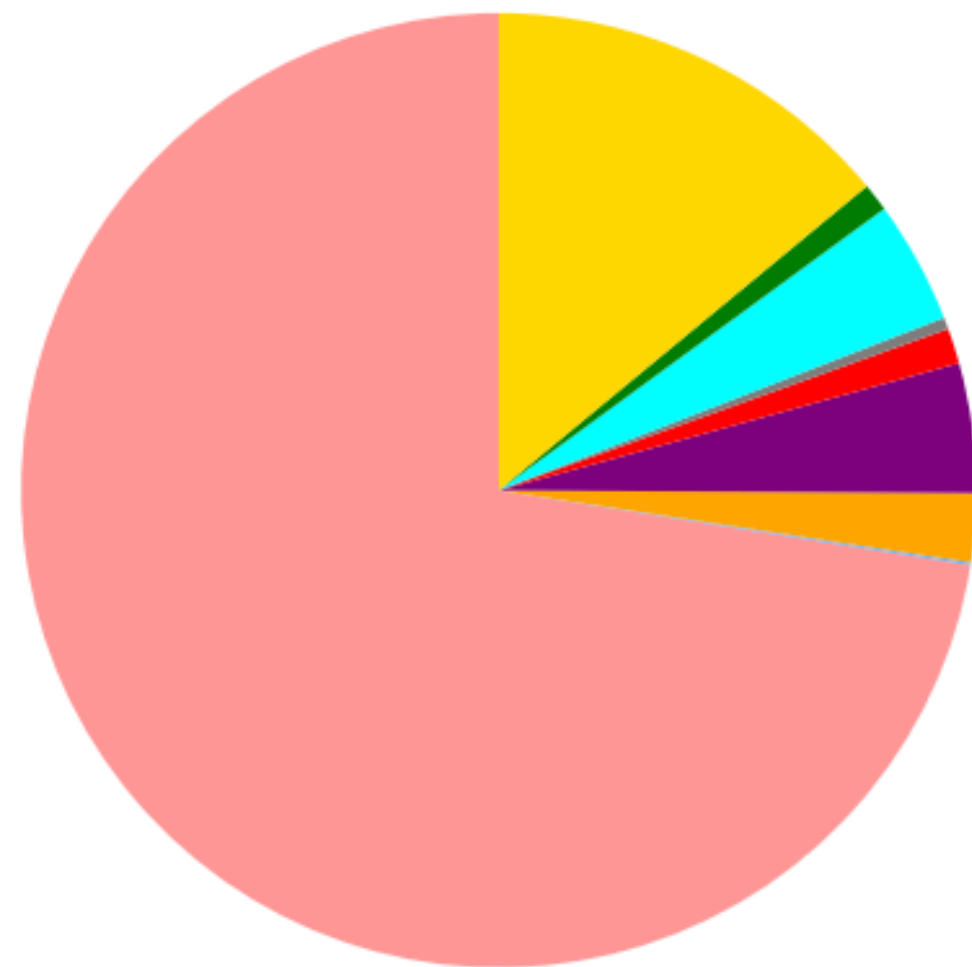
4.1 Dataset description

- Source: araCovid_sentiment.tsv
- Size: 4,128 unique Arabic comments

[illegible]

4.2 Data Cleaning

Distribution of the Data



Arabic Words: 72.50%

Latin Words: 0.09%

Emojis: 2.32%

Hashtags: 4.39%

Mentions: 1.20%

Digits: 0.39%

Special Characters: 4.16%

URLs: 0.95%

Stop Words: 14.00%

4.2 Data Cleaning

Column	Original Text	Cleaned Text
0	علاج فايروس الكورونا كل يوم الصبح الواحد يقرط ثلاث حبات ثوم، ومع إنو الثوم ما إله علاقة بفايروس الكورونا بس هكذا بتضمن لا حدا يقرب عليك ولا حدا يبوسك. 🤪🤪🤪 الله يبعد مننا ومنكم البلاء يارب 🤪	علاج ايروس الكورونا كل يوم الصبح الواحد يقرط ثلاث حبات ثوم ومع نو الثوم ما له علاقة بايروس الكورونا بس هكذا بتضمن لا حدا يقرب عليك ولا حدا يبوسك الله يبعد مننا ومنكم البلاء يارب
4	هههههههه انتظر بعد ما تخلص كورونا @Waleed_Albashaa عادي 🤪	هههههههه انتظر بعد ما تخلص كورونا عادي
10	هههه بعد كورونا إن شاء الله اما حالياً @hamedalhinai02 ملتزمين بالتباعد 🤪👍	هههه بعد كورونا ن شا الله اما حالياً ملتزمين بالتباعد
20	هههههههههههه بعد كورونا ن شاء الله @MaiBadran4	هههههههههههه بعد كورونا ن شا الله
30	من كتر أخبار أعداد الموتى بسبب القتل والحرب و الكورونا . مذیعة قناة العربية غلطت و قالت :- {{{ سنموت بعد الفاصل }}} 🤪 https://t.co/5kYelqyJ0k	من كتر أخبار أعداد الموتى بسبب القتل والحرب و الكورونا مذیعة قناة العربية غلطت و قالت سنموت بعد الفاصل

4.3 Tokenization

- Split each comment into individual words (tokens) using NLTK
- Store tokens in a new column for further processing

	Text	cleaned_text	tokenized_text
0	علاج قايروس الكورونا* كل يوم الصبح الواحد يقرط* ثلاث حبات ثوم، ومع إنو الثوم ما إله علاقة بقايروس الكورونا بس هكذا بتضمن لا حدا يقرب عليك ولا حدا يبوسك. 🤔🤔🤔🤔🤔🤔🤔 الله يبعد مننا ومنكم البلاء يارب	علاج ايروس الكورونا يوم الصبح الواحد يقرط ثلاث حبات ثوم ومع نو الثوم علاقة بايروس الكورونا بتضمن حدا يقرب حدا يبوسك الله يبعد مننا ومنكم البلا يارب	[علاج, ايروس, الكورونا, يوم, الصبح, الواحد], يقرط, ثلاث, حبات, ثوم, ومع, نو, الثوم, علاقة, بايروس, الكورونا, بتضمن, حدا, يقرب, حدا, يبوسك, الله, يبعد, مننا, ومنكم, البلاء, يارب]
4	ههههههه انتظر بعد ما تخلص @Waleed_Albashaa كورونا عادي 🤔	ههههههه انتظر تخلص كورونا عادي	[ههههههه, انتظر, تخلص, كورونا, عادي]
10	هههه بعد كورونا إن شاء الله اما @hamedalhinai02 حاليا ملتزمين بالتباعد 🤔👍	هههه كورونا شا الله اما حاليا ملتزمين بالتباعد	[هههه, كورونا, شا, الله, اما, حاليا, ملتزمين], بالتباعد]

4.4 Lable Encoding

For ML

- Convert sentiment labels from text to integers:
 - Negative $\rightarrow 0$
 - Neutral $\rightarrow 1$
 - Positive $\rightarrow 2$

For DL

- the integer-encoded labels were further converted into one-hot vectors:
 - $0 \rightarrow [1, 0, 0]$
 - $1 \rightarrow [0, 1, 0]$
 - $2 \rightarrow [0, 0, 1]$

5. Methodology

1

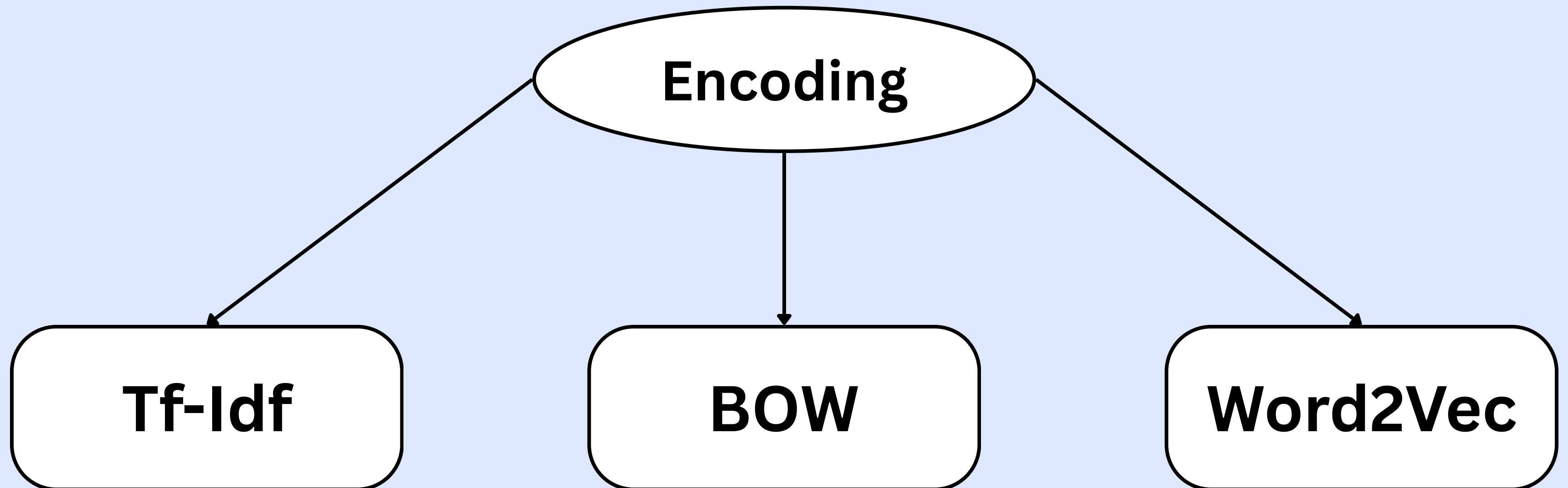
Feature extraction

2

Models

5.1 feature extraction

Feature extraction transforms raw text into numerical representations suitable for machine learning models and deep learning.



1.TF-IDF (Term Frequency-Inverse Document Frequency)

Assigns weights to words based on their frequency in a document relative to their frequency across all documents, highlighting important terms.

	يلعن	يروح	يارب	كورونا	فيروس	شي	ان	الوبا	الموت	اللي	اللهم	الله	الكورونا	الصين	الصبر
0	0.0	0.0	0.591871	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.27518	0.757605	0.0	0.0
1	0.0	0.0	0.000000	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.000000	0.000000	0.0	0.0
2	0.0	0.0	0.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.000000	1.000000	0.0	0.0
3	0.0	0.0	0.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.000000	0.000000	0.0	0.0
4	0.0	0.0	0.000000	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.000000	0.000000	0.0	0.0

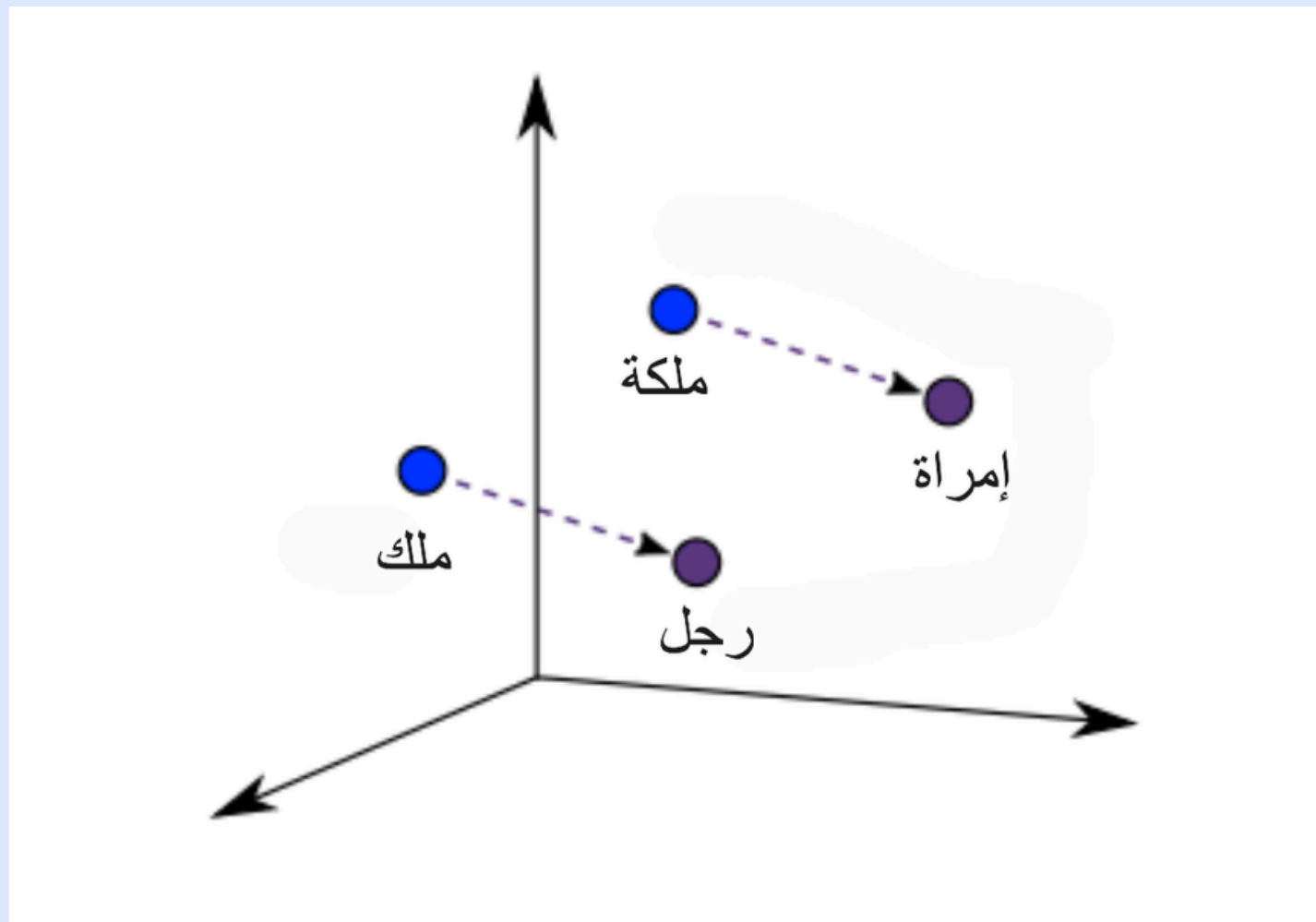
2.BOW (Bag Of Words)

Represents text by counting the occurrence of each word in the vocabulary, disregarding grammar and word order.

	يلعن	يروح	يارب	كورونا	فيروس	شي	ان	الوبا	الموت	اللي	اللهم	الله	الكورونا	الصين	الصبر
0	0	0	1	0	0	0	0	0	0	0	0	1	2	0	0
1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

3.W2V(Word to Vector)

Uses pre-trained AraVec embeddings to map Arabic words into dense vectors that capture semantic and syntactic relationships.



ملك: [... ,0.56 ,0.34- ,0.21]

رجل: [... ,0.49 ,0.29- ,0.18]

ملكة: [... ,0.60 ,0.31- ,0.23]

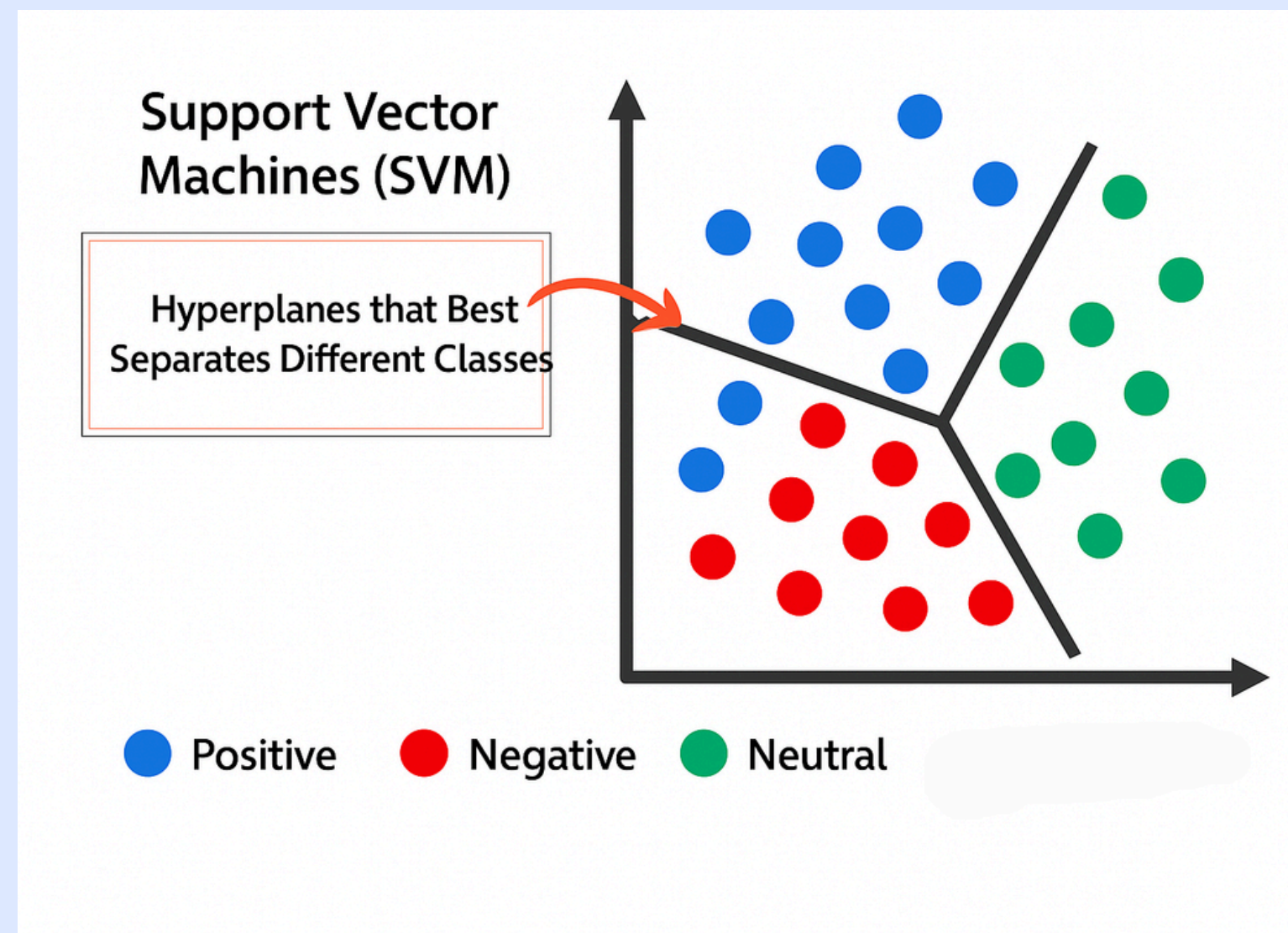
امراة: [... ,0.52 ,0.27- ,0.20]

$$\text{vector("ملك")} - \text{vector("رجل")} + \text{vector("امراة")} \approx \text{vector("ملكة")}$$

5.2 Models

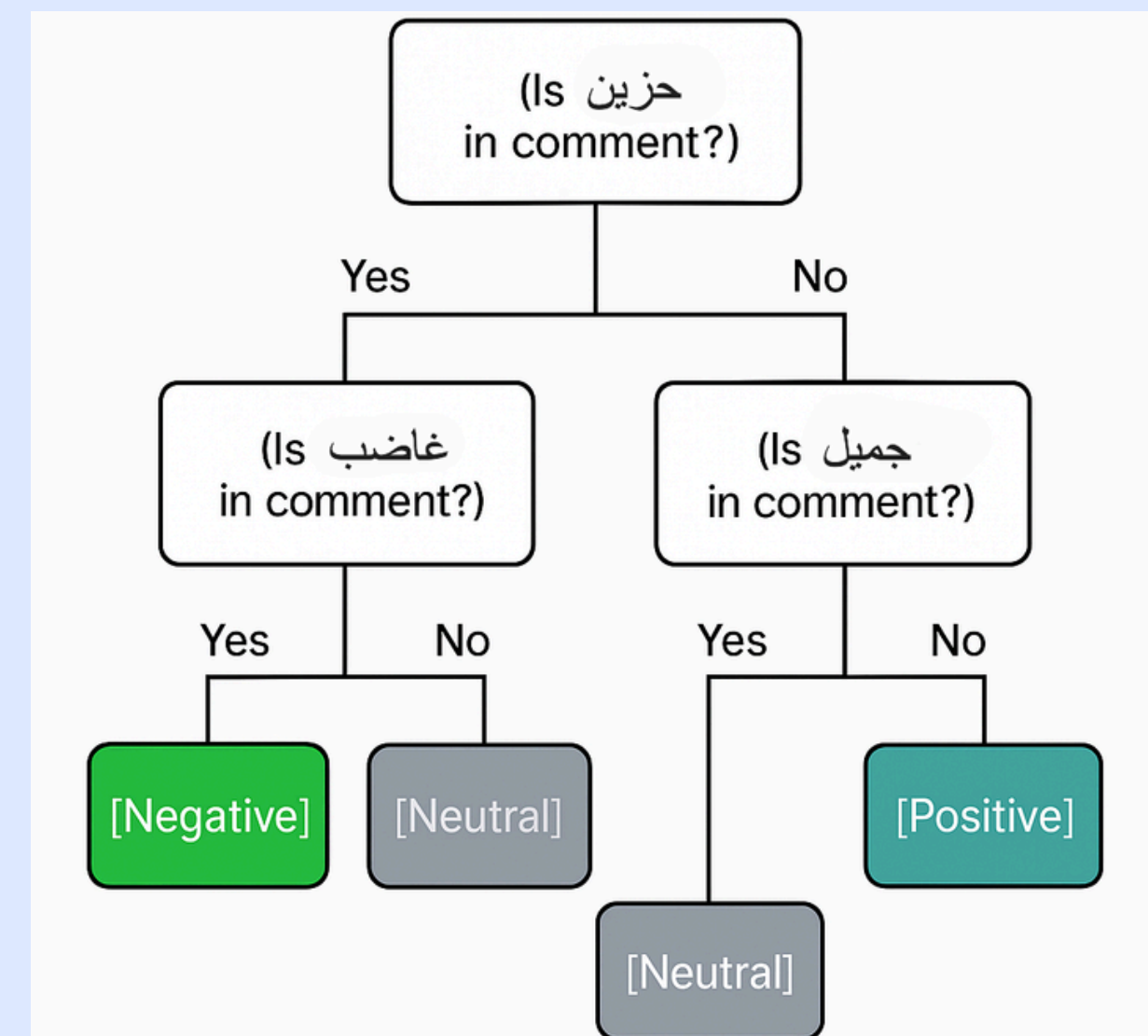
SVM(Suport Vector Machine)

SVM is known for its ability to find the best hyperplane that separates classes. It performs well in high-dimensional spaces, which makes it suitable for text data represented by TF-IDF or Bag of Words.



DT(Decision Tree)

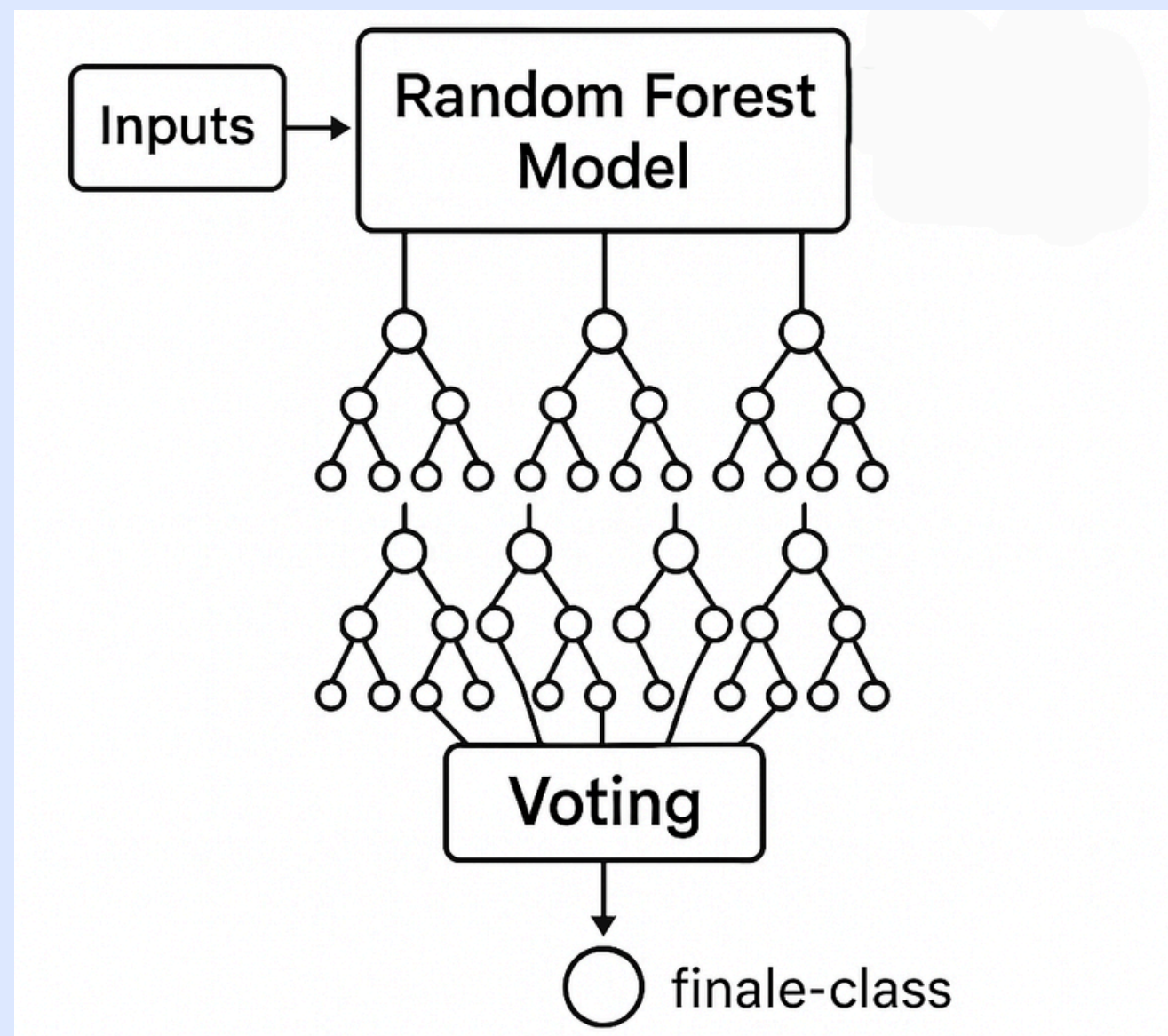
Decision Tree is a simple model. It makes decisions by splitting the data based on features. While it is easy to understand and visualize, it often suffers from overfitting, especially with noisy text data.



5.2 Models

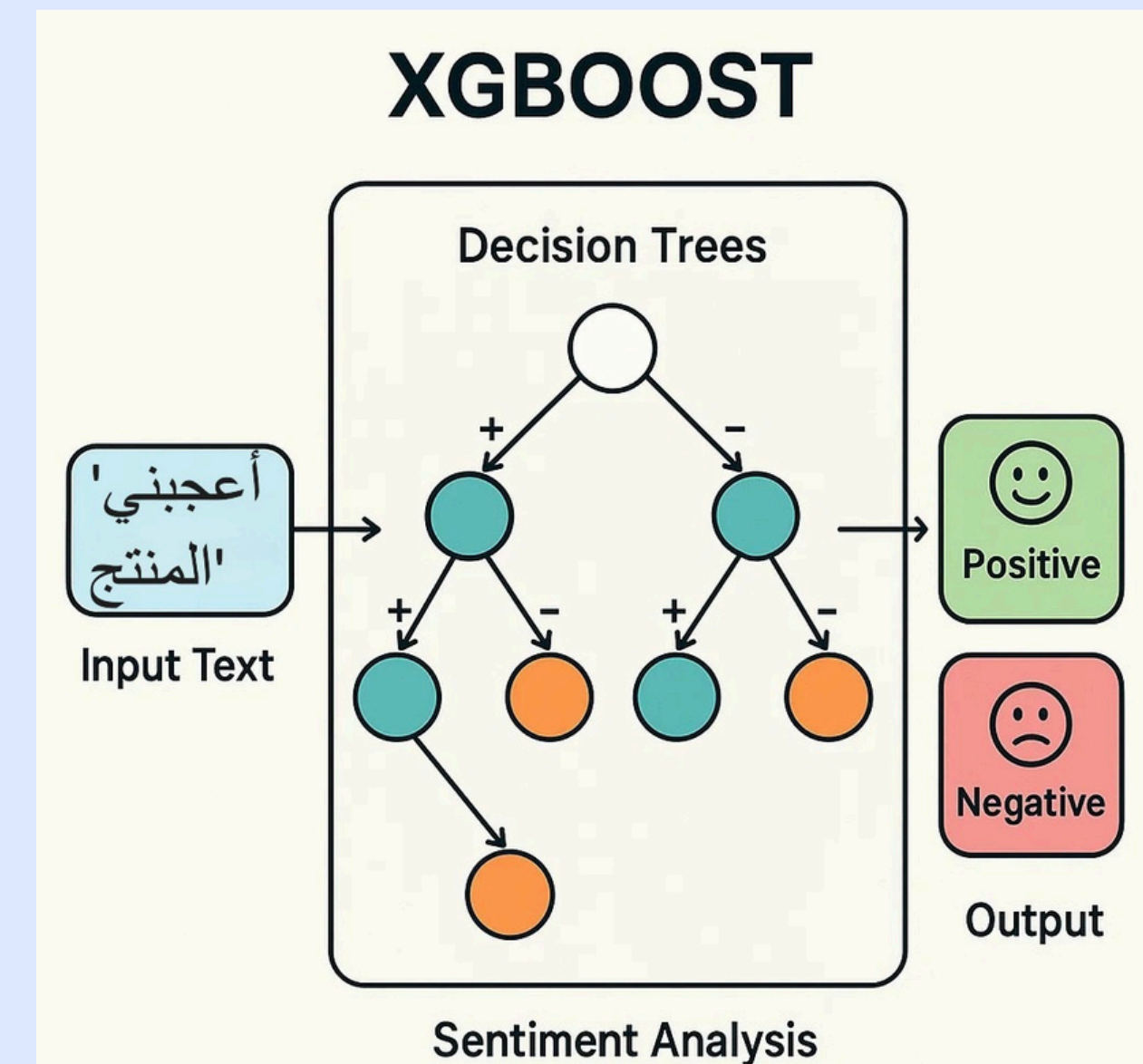
RF(Random Forest)

Random Forest is an ensemble method based on decision trees. It builds multiple trees on random subsets and combines their outputs, which helps reduce overfitting and improves performance.



XGBOOST(eXtreme Gradient Boosting)

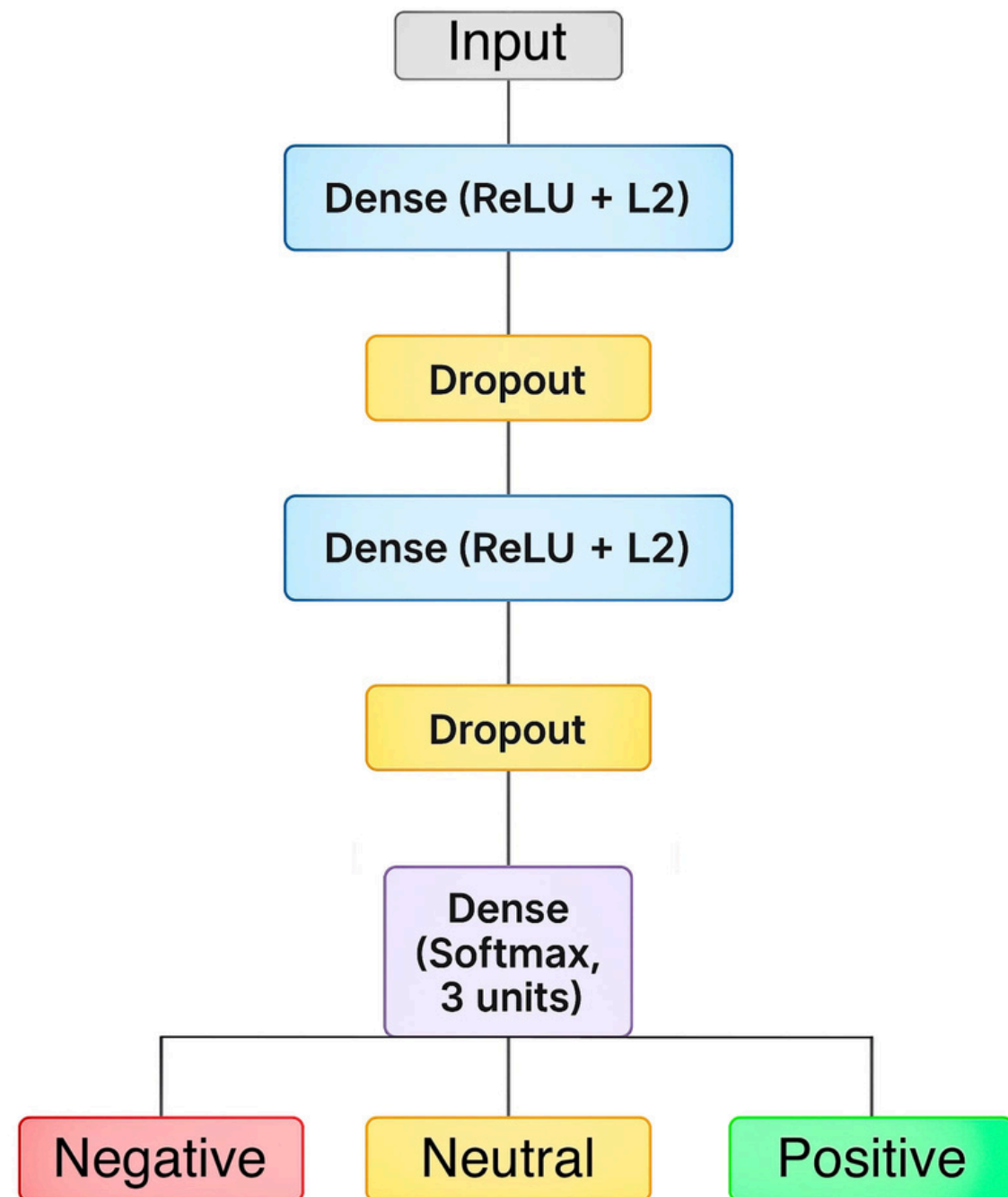
XGBoost is a boosting algorithm that builds trees sequentially, where each new tree corrects the errors of the previous ones. It is known for its speed and high performance in classification problems.



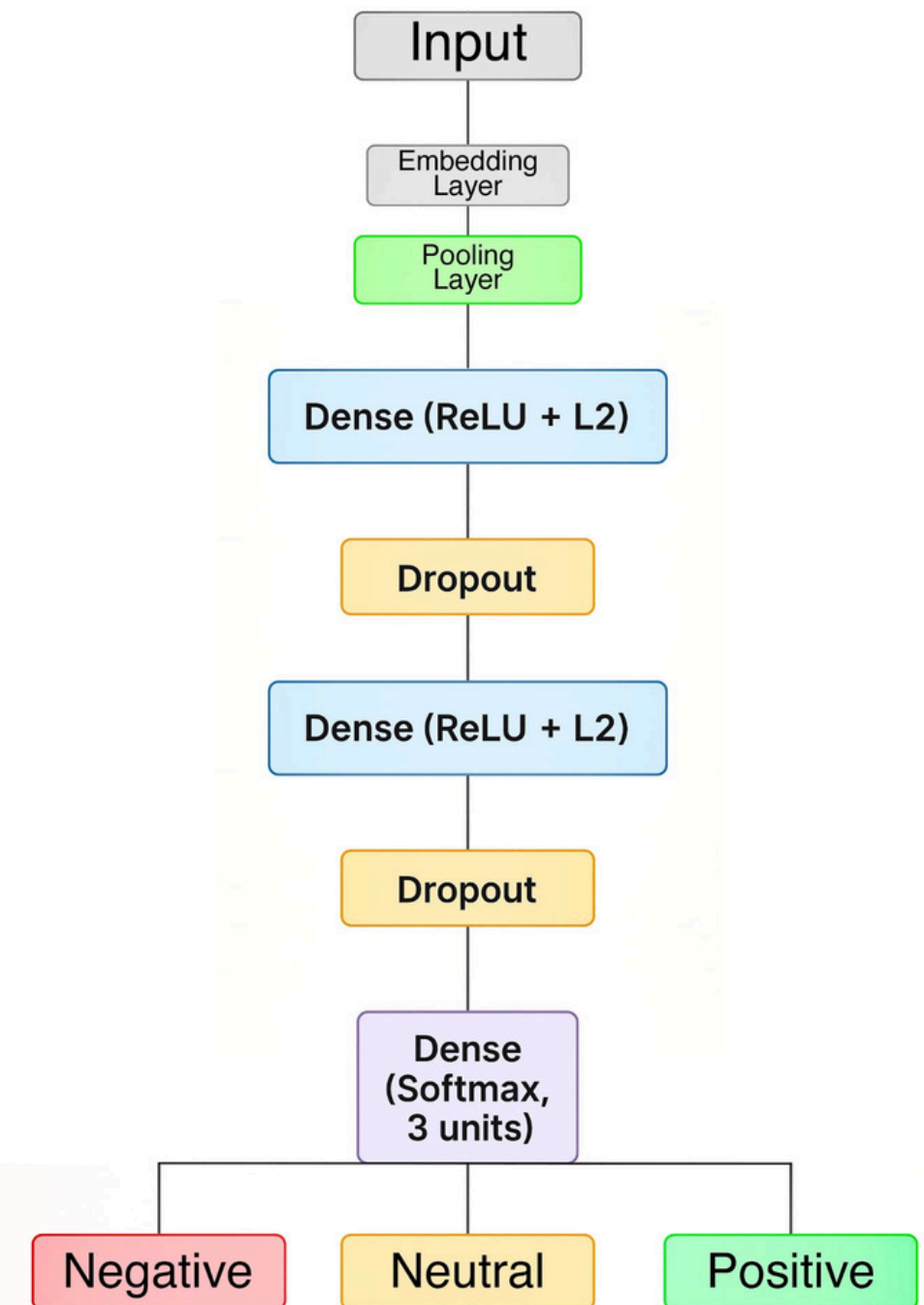
5.2 Models

For DL

NN (neural Network) for TF-IDF & BOW



NN (neural Network) for W2V

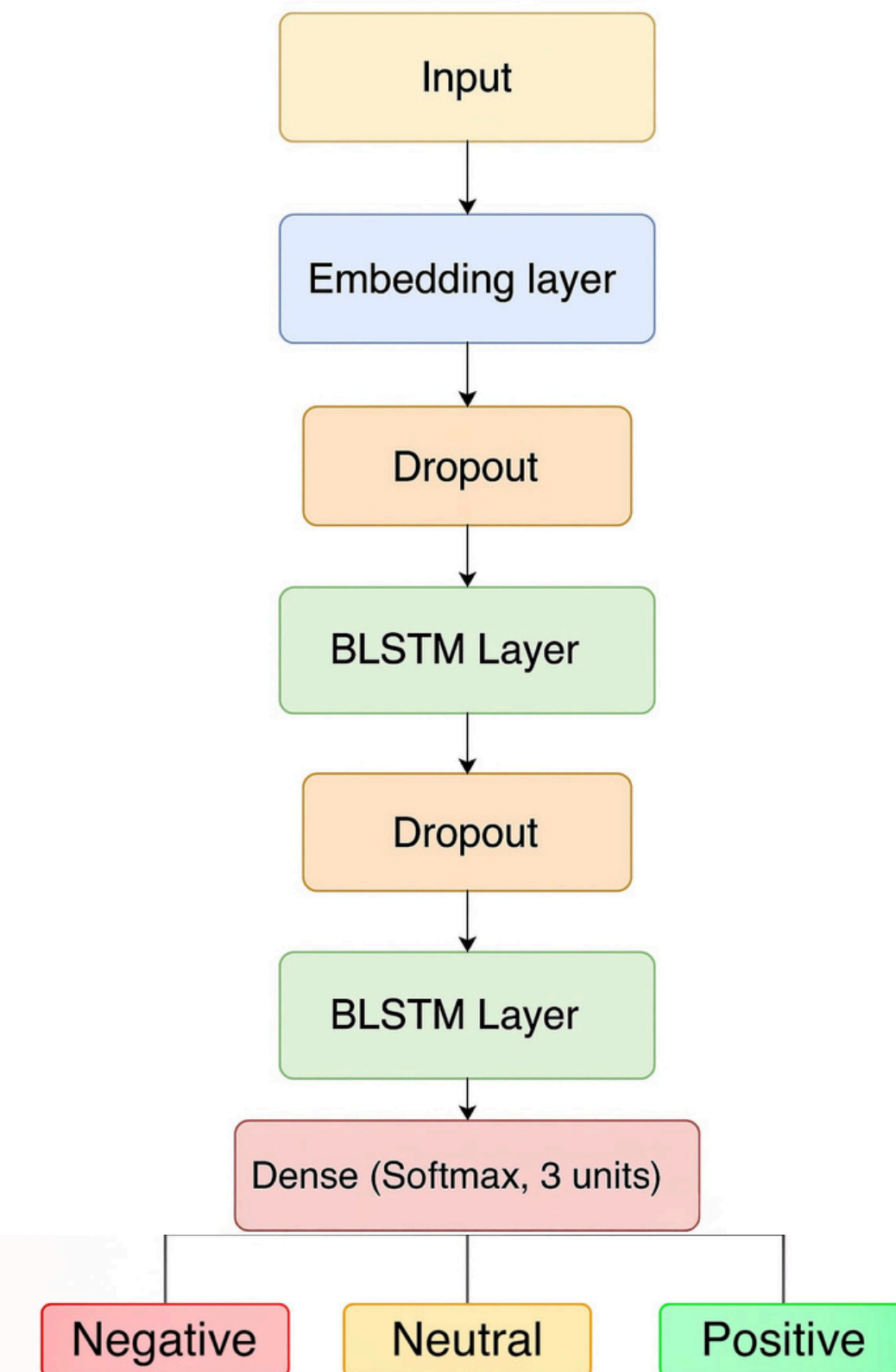


5.2 Models

For DL

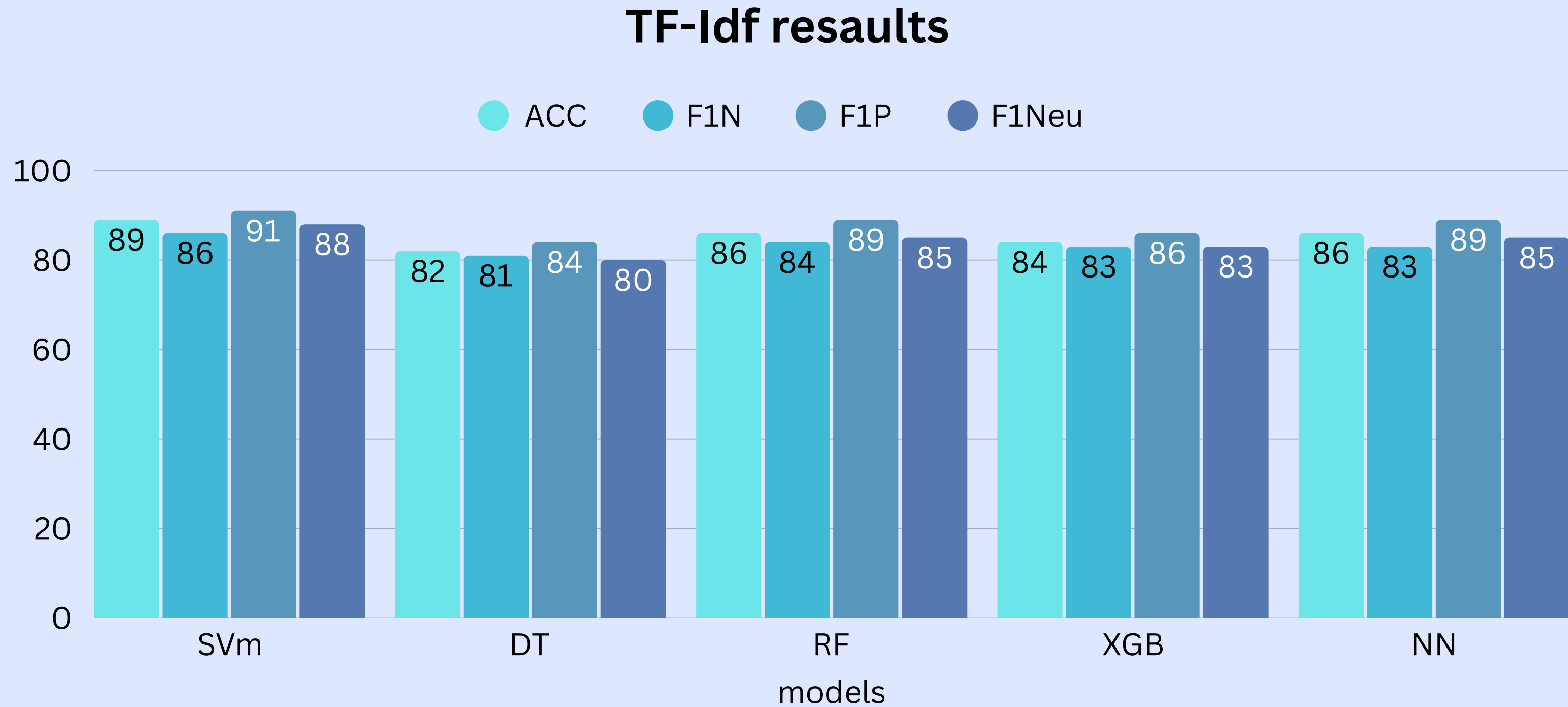
B-LSTM (Bidirectional Long Short-Term Memory) for W2V

To better capture the context and word order in Arabic sentences, we used a Bidirectional Long Short-Term Memory network. This model processes the sentence in both directions, which is important for languages like Arabic where word meaning can depend on both previous and next words



6-Results & Analysis

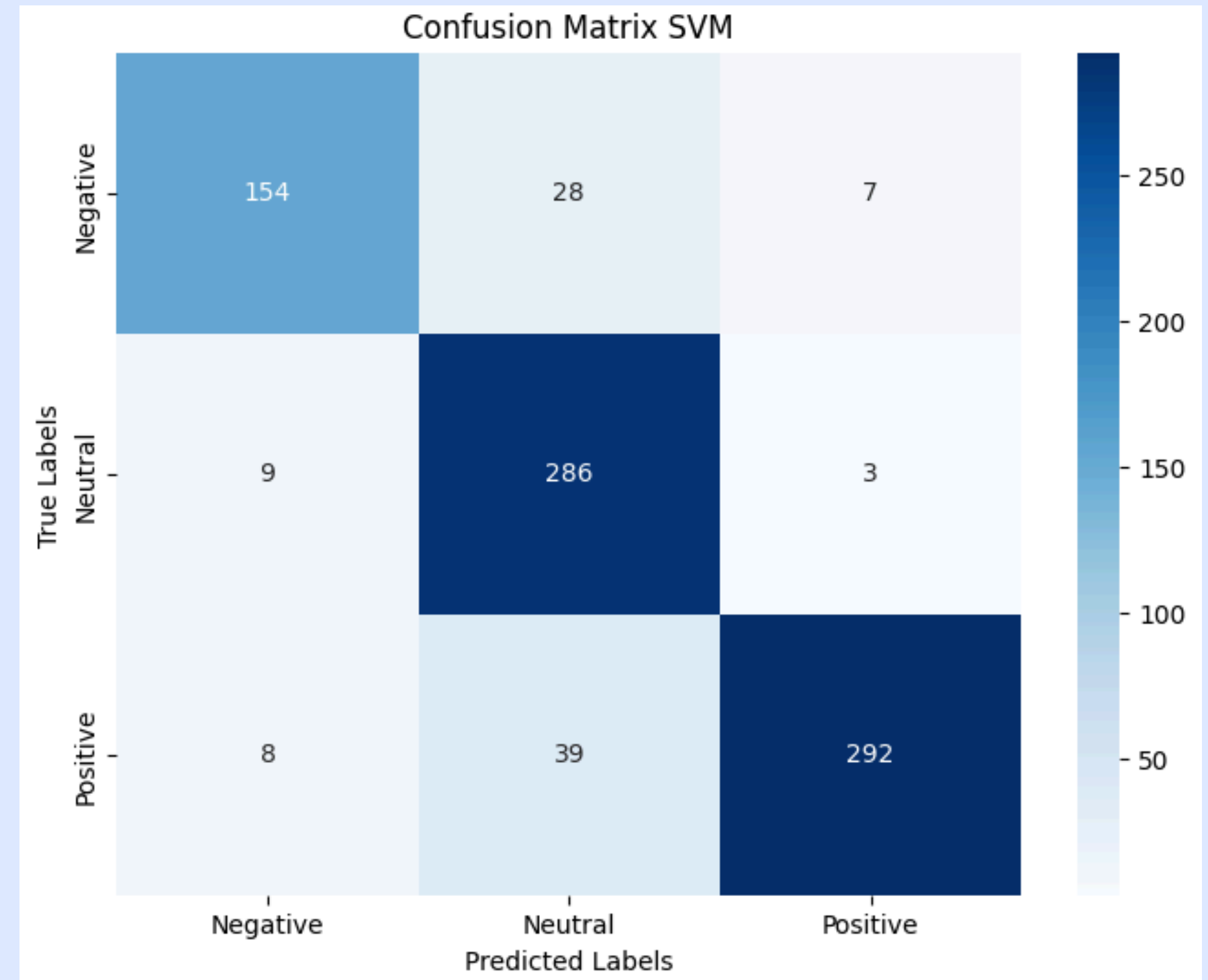
- **TF-IDF:**



6-Results & Analysis

- **SVM:**

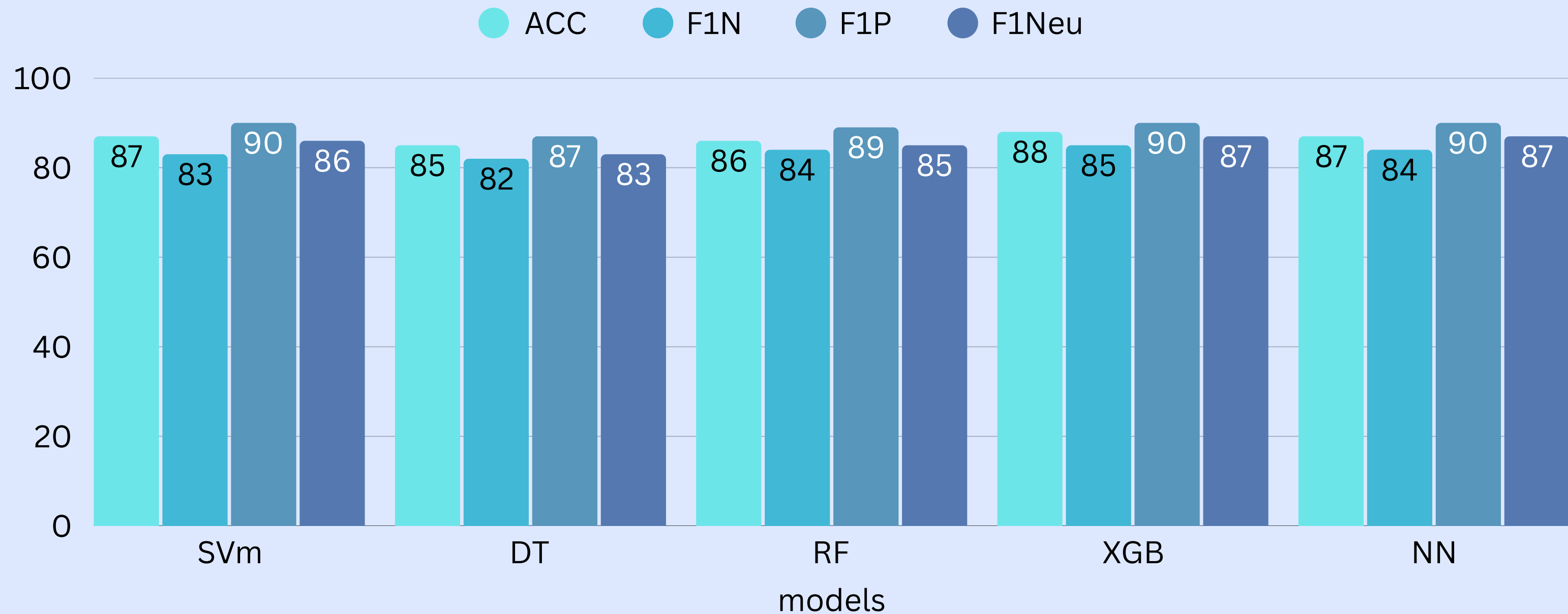
The SVM confusion matrix shows most predictions are correct, with few misclassifications mainly between neutral and negative or positive and neutral classes.



6-Results & Analysis

- Bag of Words:

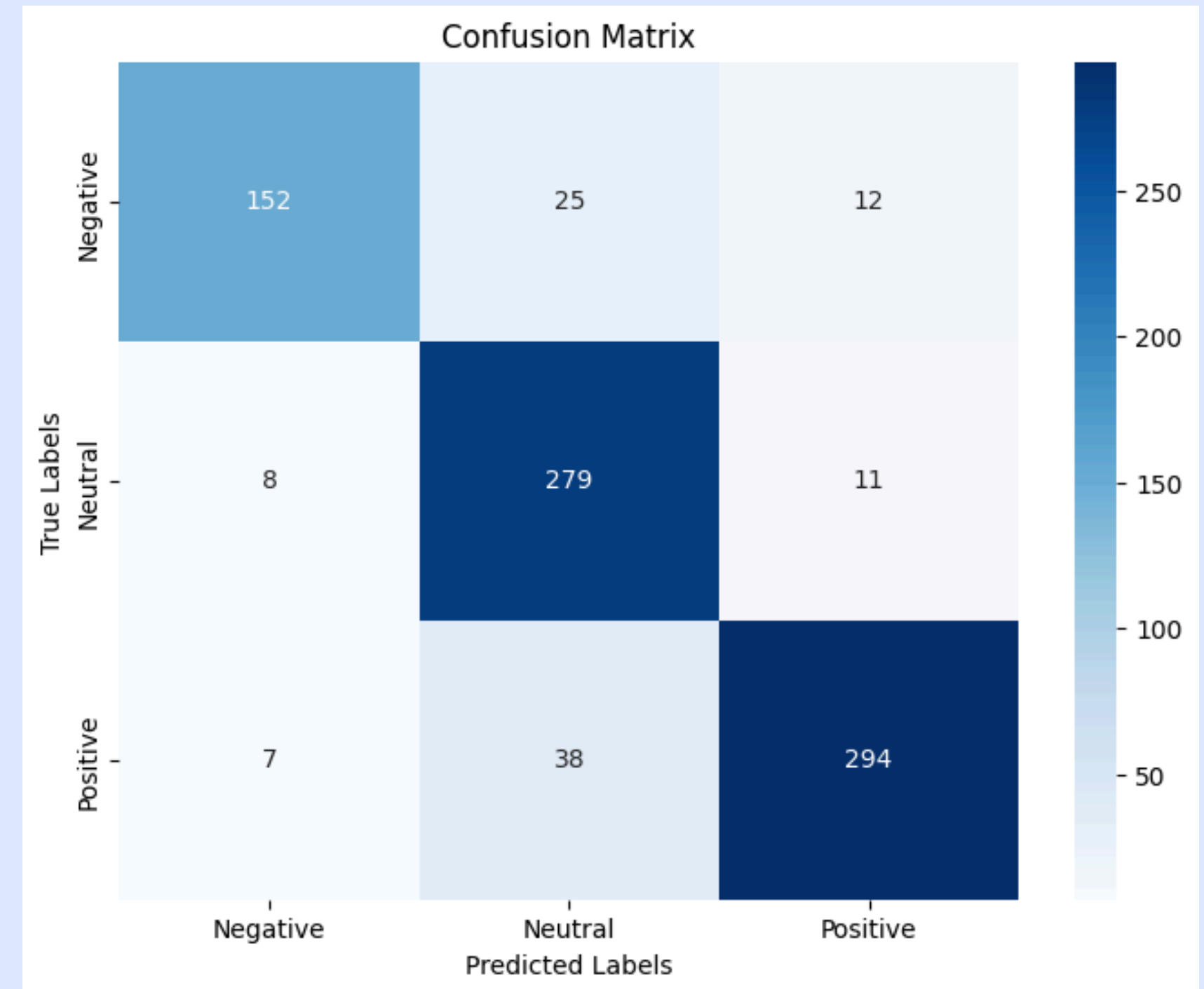
BOW results



6-Results & Analysis

- **XGBOOST:**

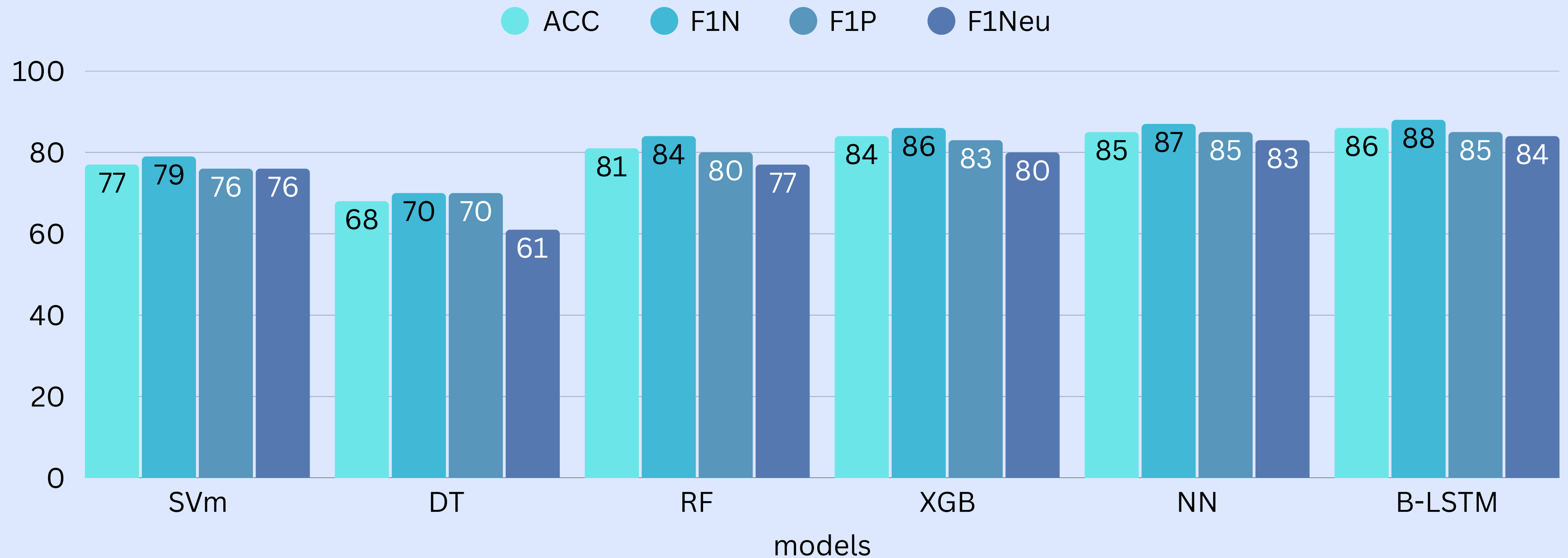
The XGBoost confusion matrix shows strong alignment with true labels, with most errors involving confusion between neutral and the other classes especially neutral being misclassified as positive.



6-Results & Analysis

- Word2Vec:

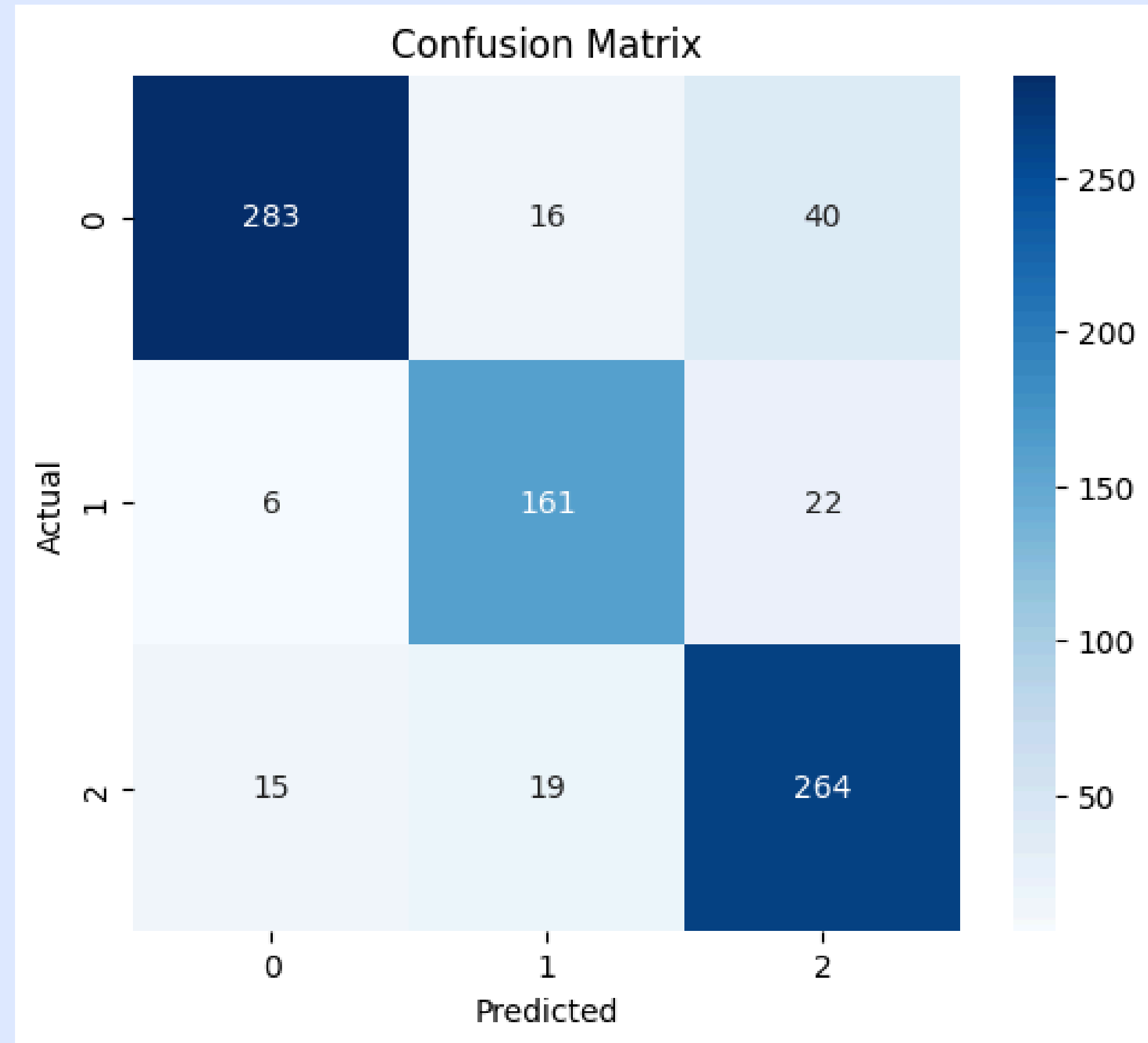
W2V results



6-Results & Analysis

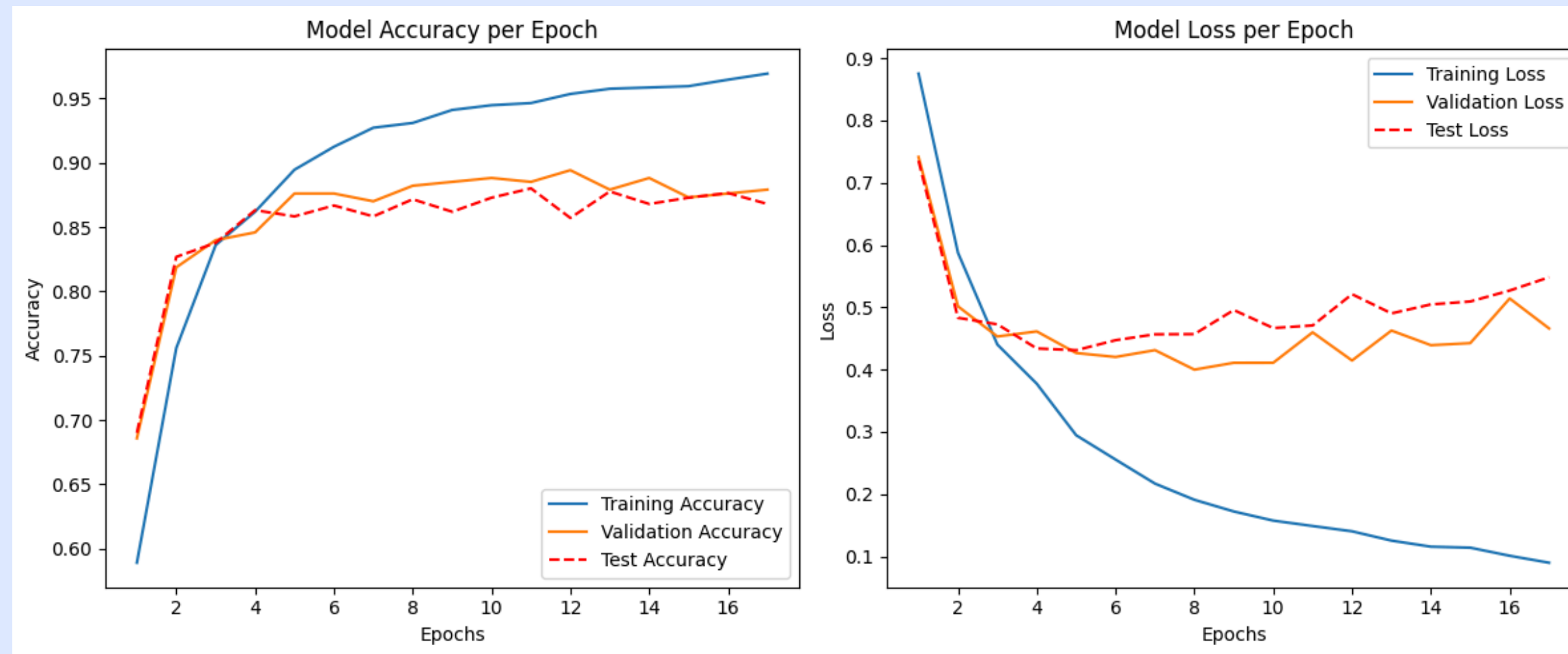
- **B-LSTM:**

The confusion matrix reveals that B-LSTM rarely confuses negative and positive sentiments, and handles neutral comments with greater accuracy than other models.



6-Results & Analysis

- **B-LSTM:**



The B-LSTM learning curves show rapid and stable improvement in both accuracy and loss for training, validation, and test sets indicating the model learned efficiently and generalized well without overfitting.

7 Conclusion

To conclude, this master's thesis focused on experimenting with sentiment analysis of Arabic dialects, specifically in social media contexts. We conducted a comprehensive evaluation using a wide range of text encoding techniques—namely TF-IDF, Bag of Words, and Word2Vec—and tested them across both traditional machine learning algorithms and deep learning models.

By systematically comparing these approaches, we gained valuable insights into their strengths and limitations when dealing with dialectal Arabic, which is often informal, varied, and under-resourced.

7 Conclusion

Our results showed that traditional models like SVM and XGBoost performed very well with sparse representations like TF-IDF and Bag of Words. However, the best overall performance was achieved using Word2Vec embeddings with a Bidirectional LSTM model. This combination captured deeper semantic and contextual meaning, making it more effective for classifying sentiment in dialectal Arabic.

In the end, this study not only highlights the potential of deep learning with semantic embeddings for Arabic sentiment analysis, but also emphasizes the importance of choosing the right feature representation and model combination.

8 Future Improvements

1. Use Pretrained Deep Language Models (Transformers)
2. Improve Dataset Quality and Size
3. Multimodal Sentiment Analysis

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